

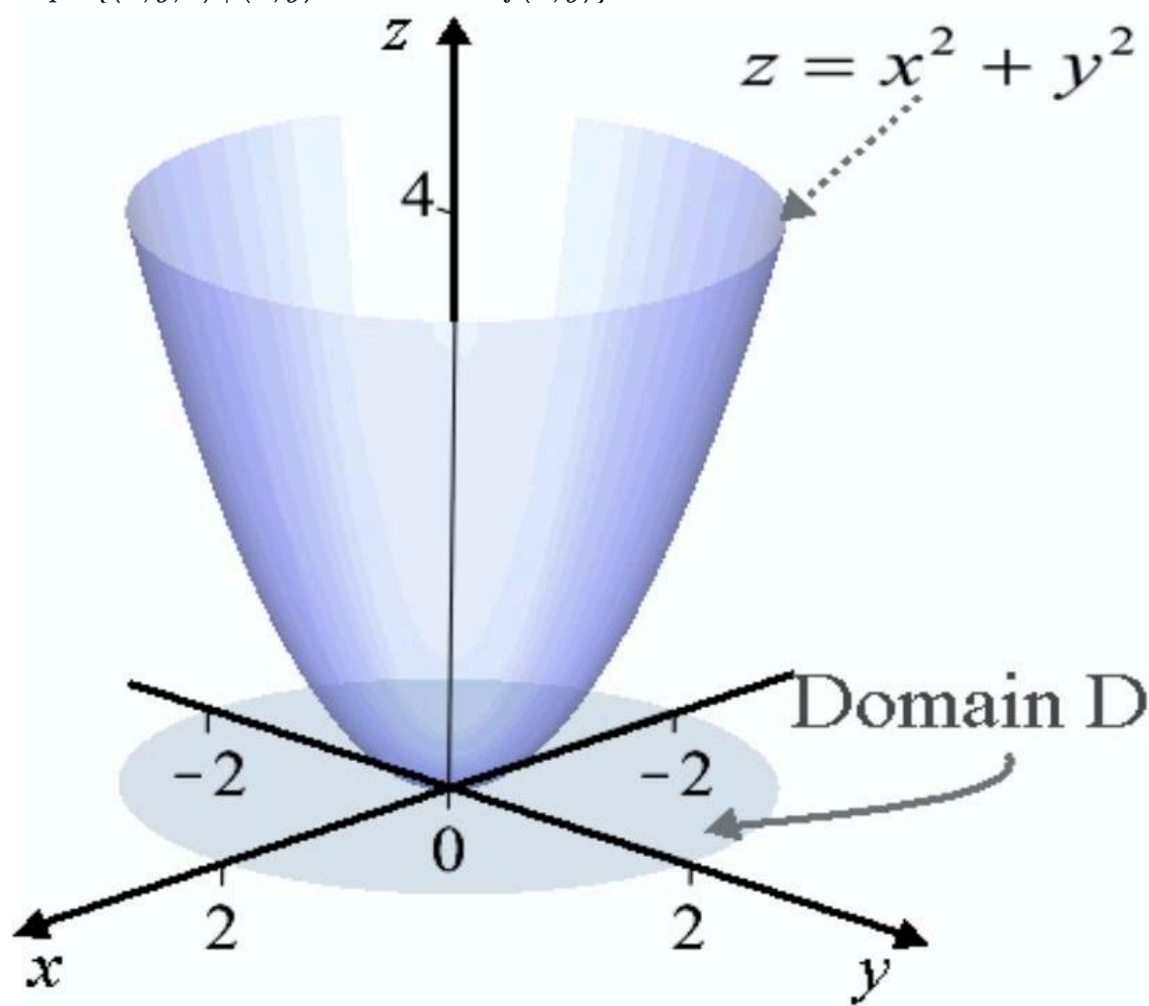
Functions of Several Variables

A function of two variables is a rule which assigns for each (x, y) in a domain $D \subset \mathbf{R}^2$ a real number $z = f(x, y)$.

A function of two variables can be given by a formula, a graph or a table.

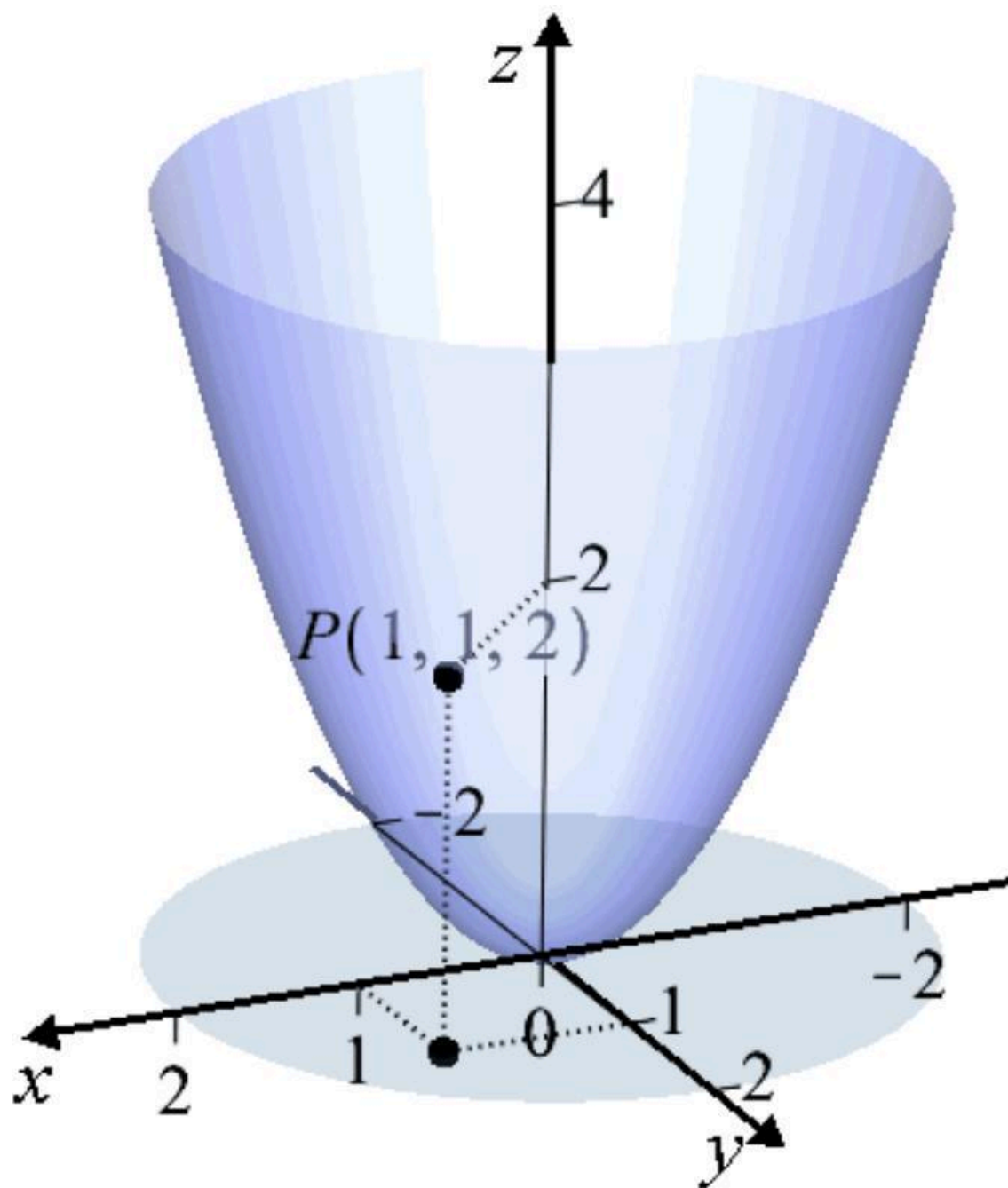
Formula: $f(x, y) = x^2 + y^2$

Graph: $\{(x, y, z) \mid (x, y) \in D \text{ and } z = f(x, y)\}$



Graph of $z = x^2 + y^2$ in the domain

$$D = \{(x, y) \mid x^2 + y^2 \leq 4\}$$



Since $f(1, 1) = 2$, the point $P(1, 1, 2)$ is on the graph of f .

Functions of Several Variables

Table: $f(x, y) = x^2 + y^2$

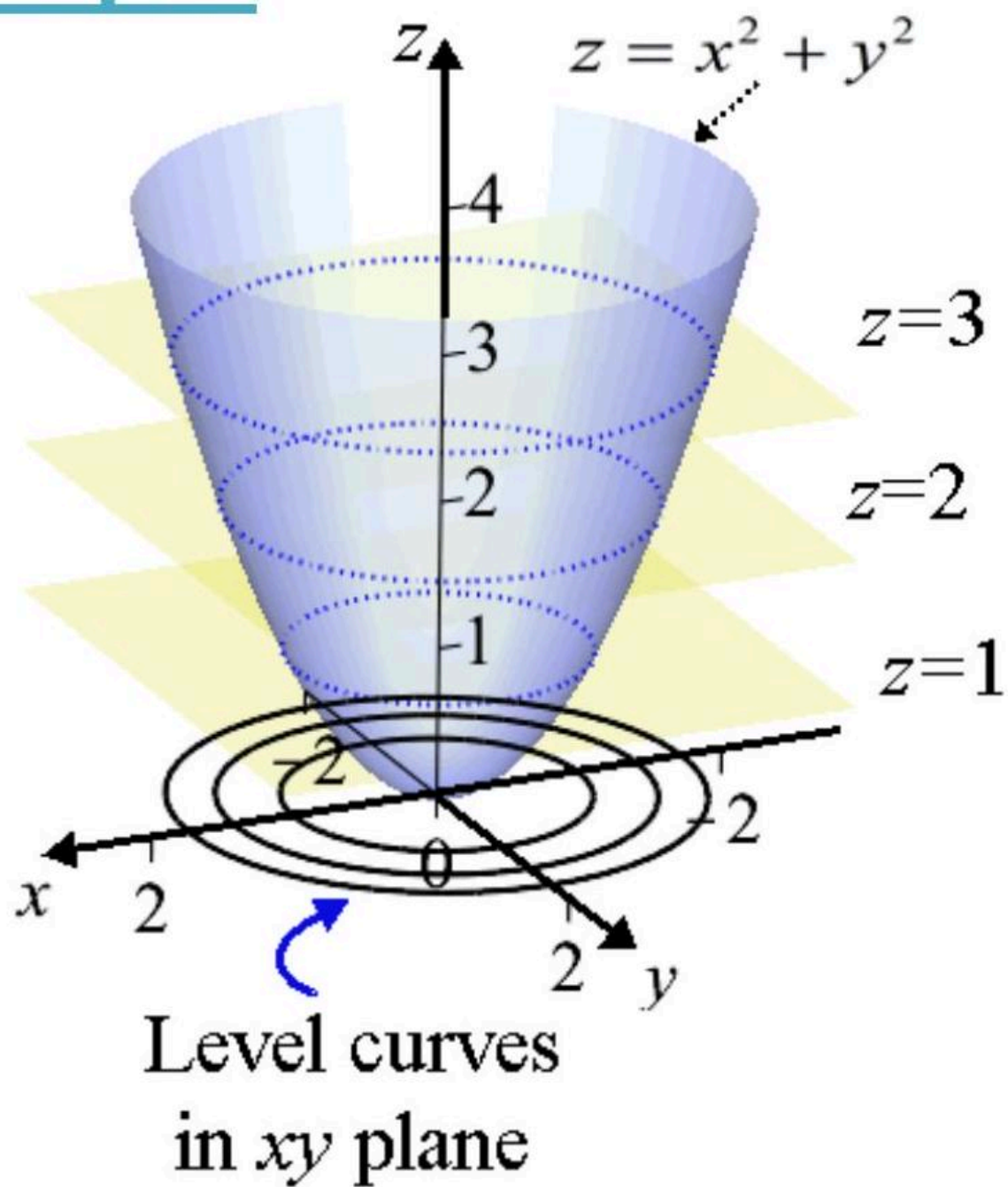
yx	1	2	3	4
−2	5	8	13	20
1	2	5	10	17
4	17	20	25	32

$$f(1, -2) = 1^2 + (-2)^2 = 5$$

$$f(2, 4) = 2^2 + 4^2 = 20$$

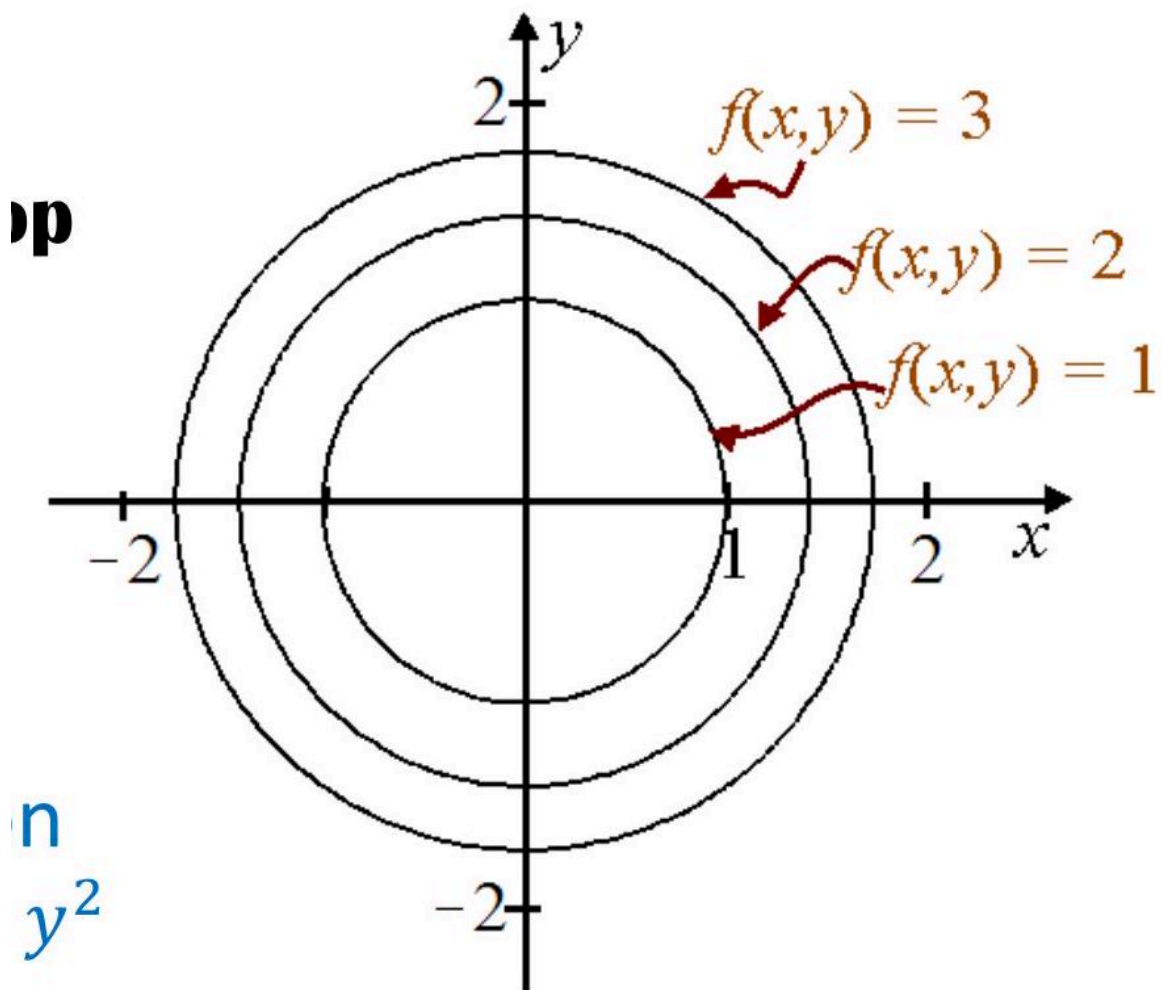
Functions of Several Variables

The Level Curves of the function are the curves with equation $f(x, y) = k$ projected down on the xy -plane. The graph of the level curves is called the contour plot



View from the top
Contour plot
of the function
 $f(x, y) = x^2 + y^2$

View from the top
Contour plot
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 $f(x, y) = x^2 + y^2$

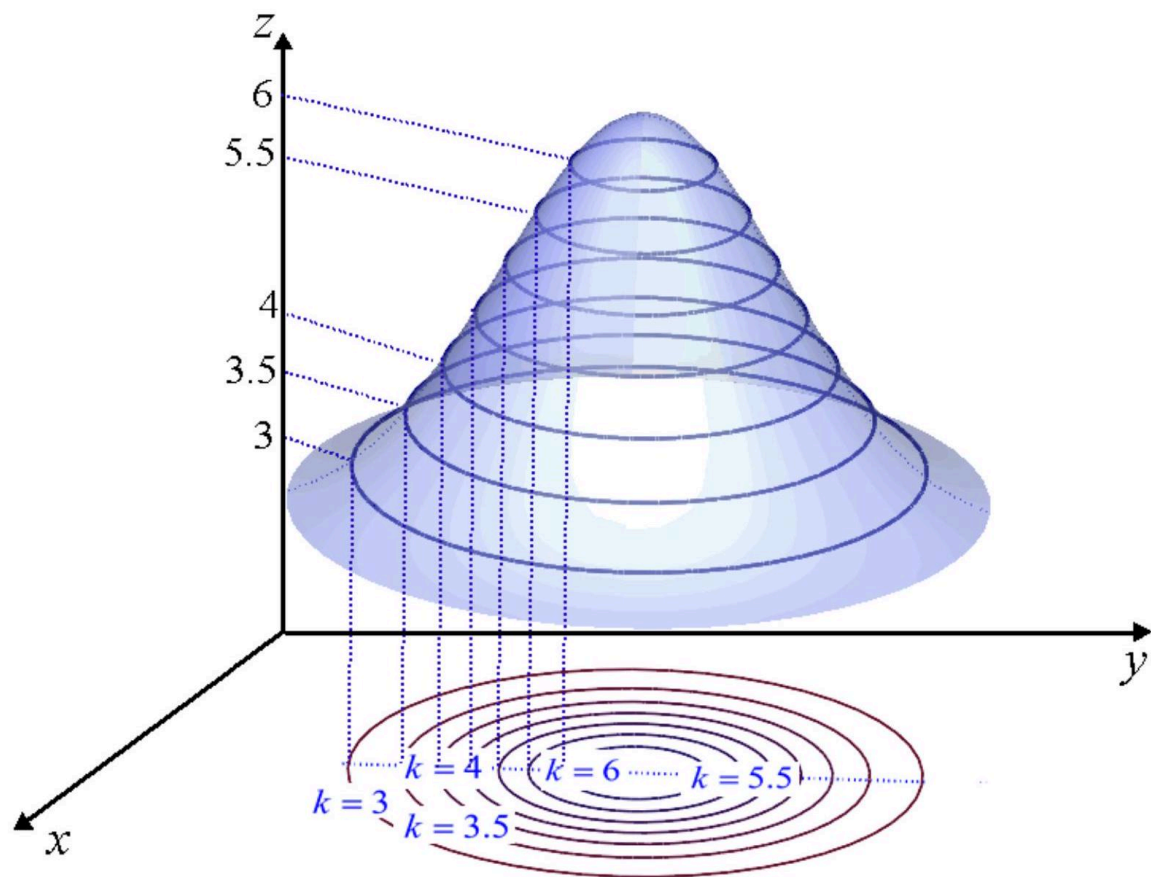


The level curves are the traces of the graph of f in the horizontal planes $z = k$ projected in the xy -plane

$k = 1 :$	circle
$k = 2 :$	$x^2 + y^2 = 1$
$f(x, y) = k -$	$k = 3 :$ circle
	$k = 2$ $x^2 + y^2 = 2$
	$x^2 + y^2 = 3$

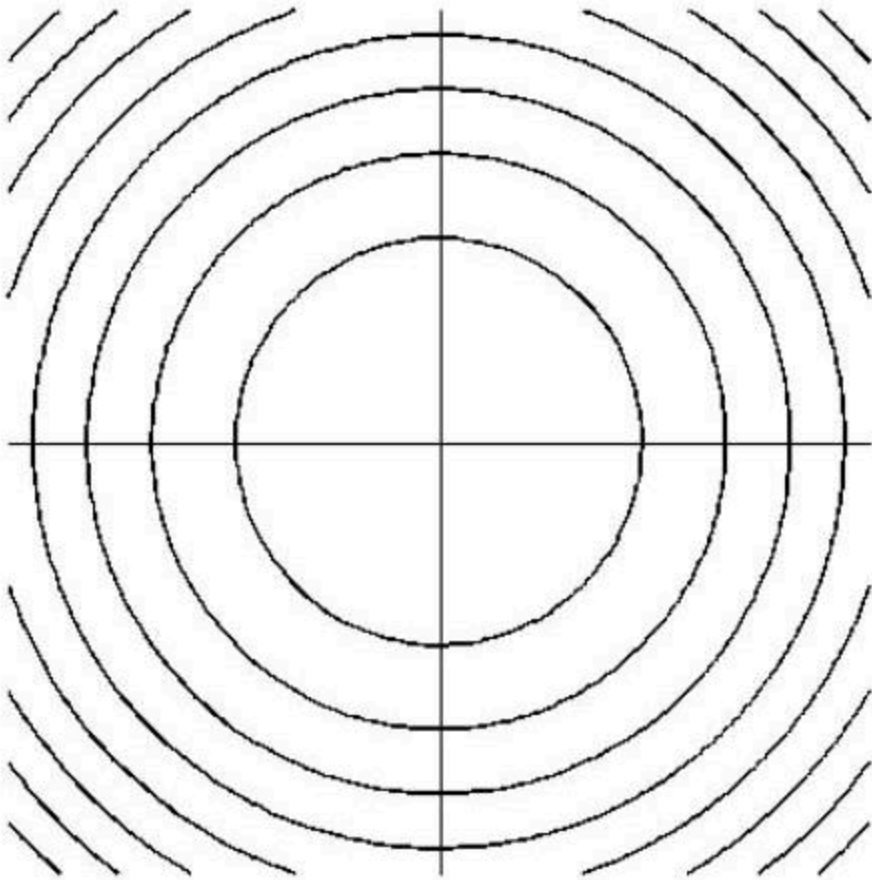
Functions of Several Variables

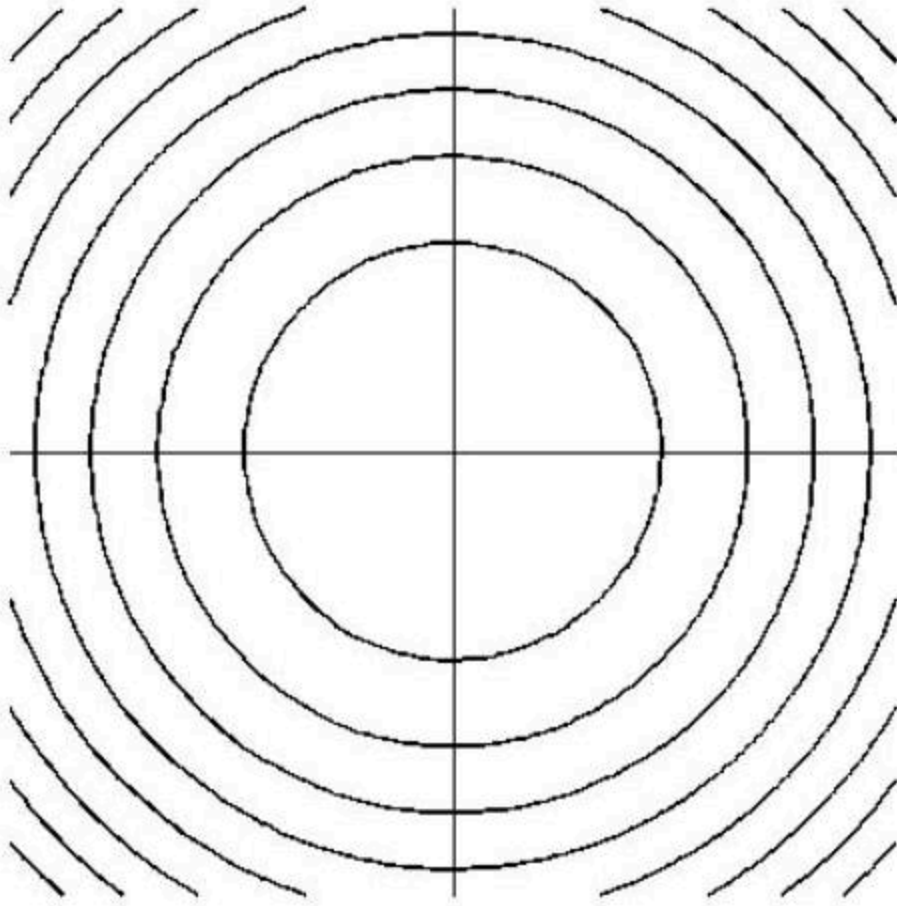
Example of surface $z = f(x, y)$ with traces on the planes $z = k$ projected down in the xy -plane.

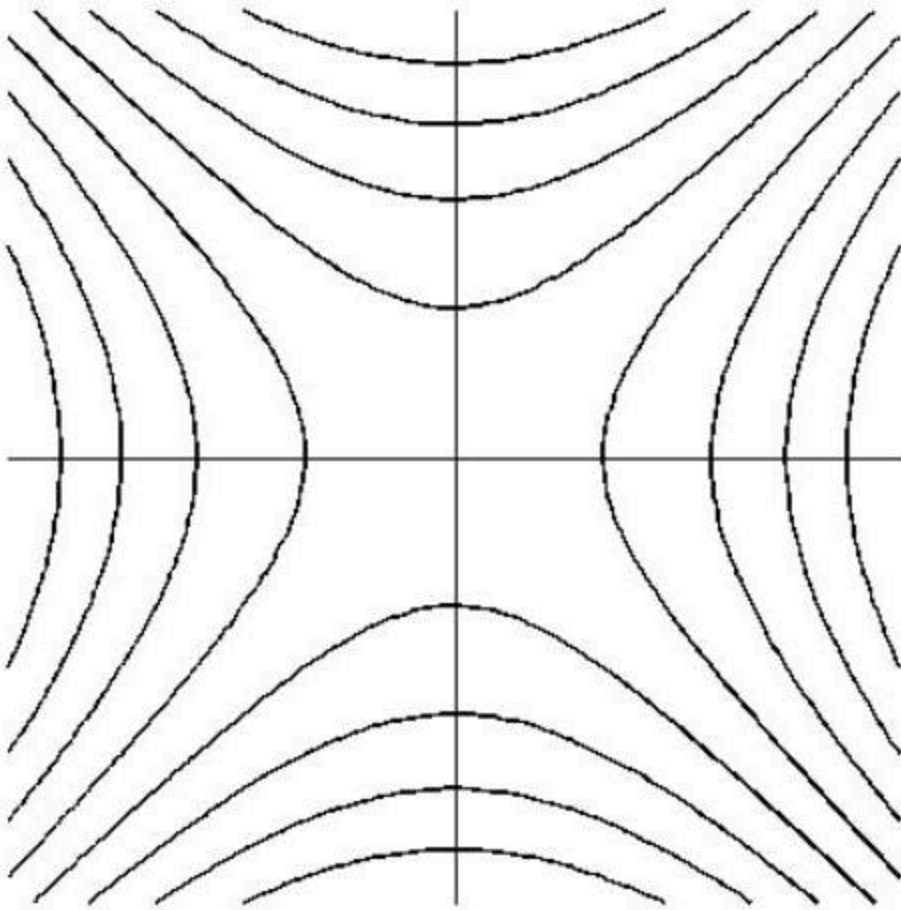


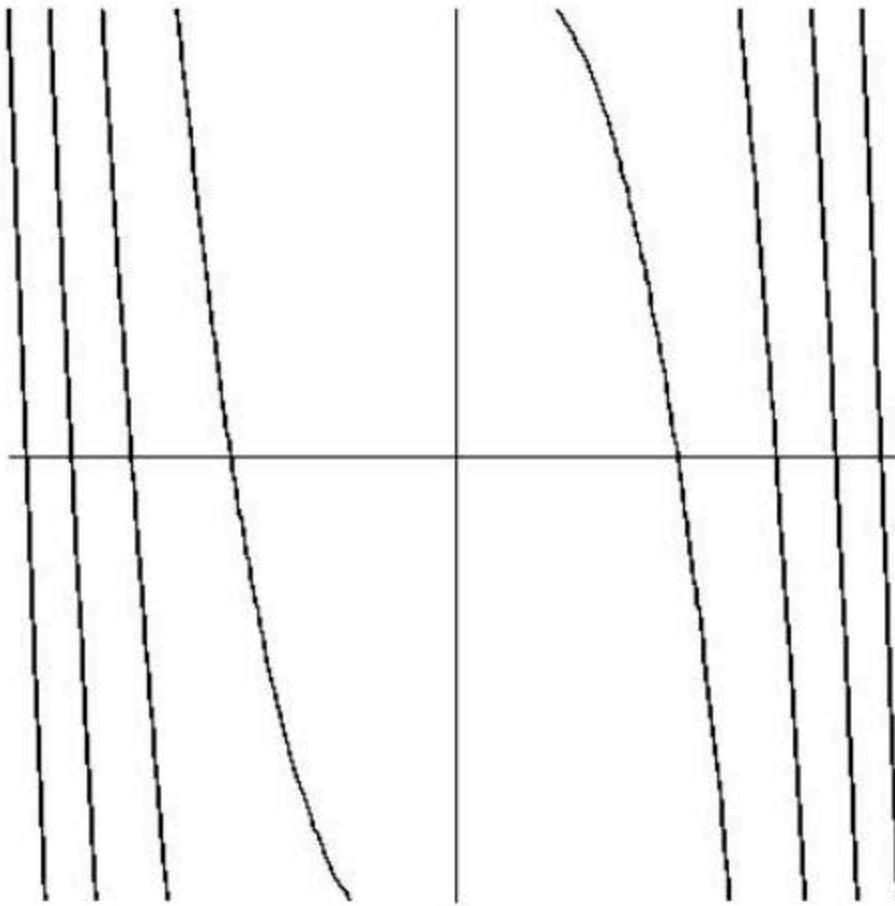
Functions of Several Variables

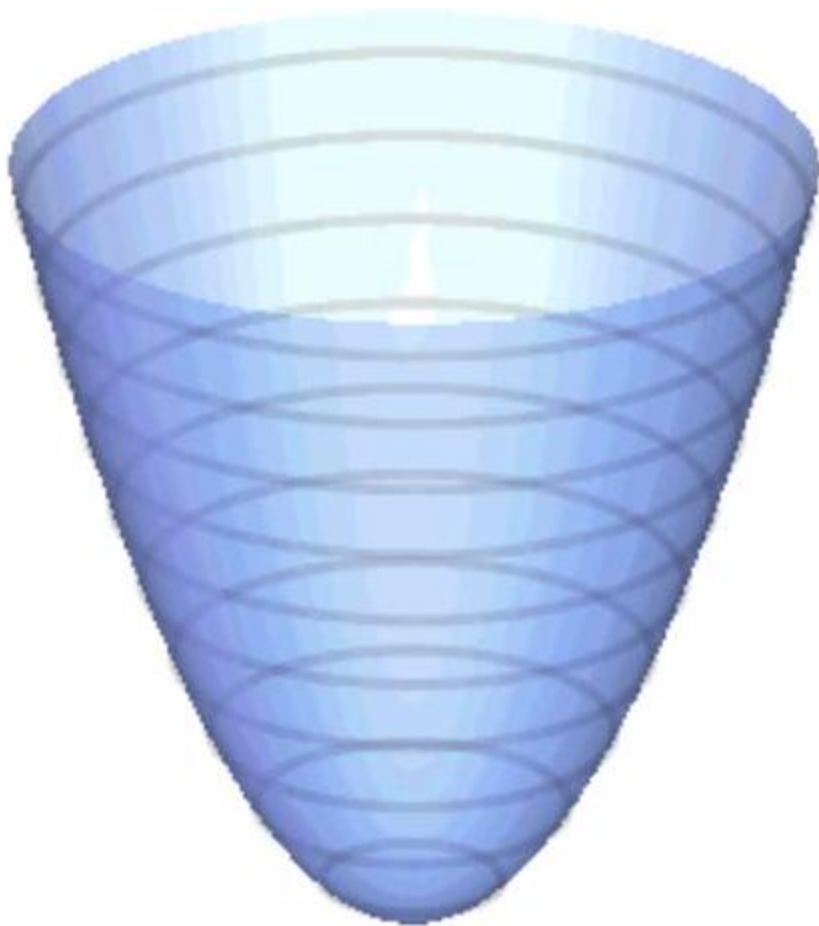
More Examples of contour plots:



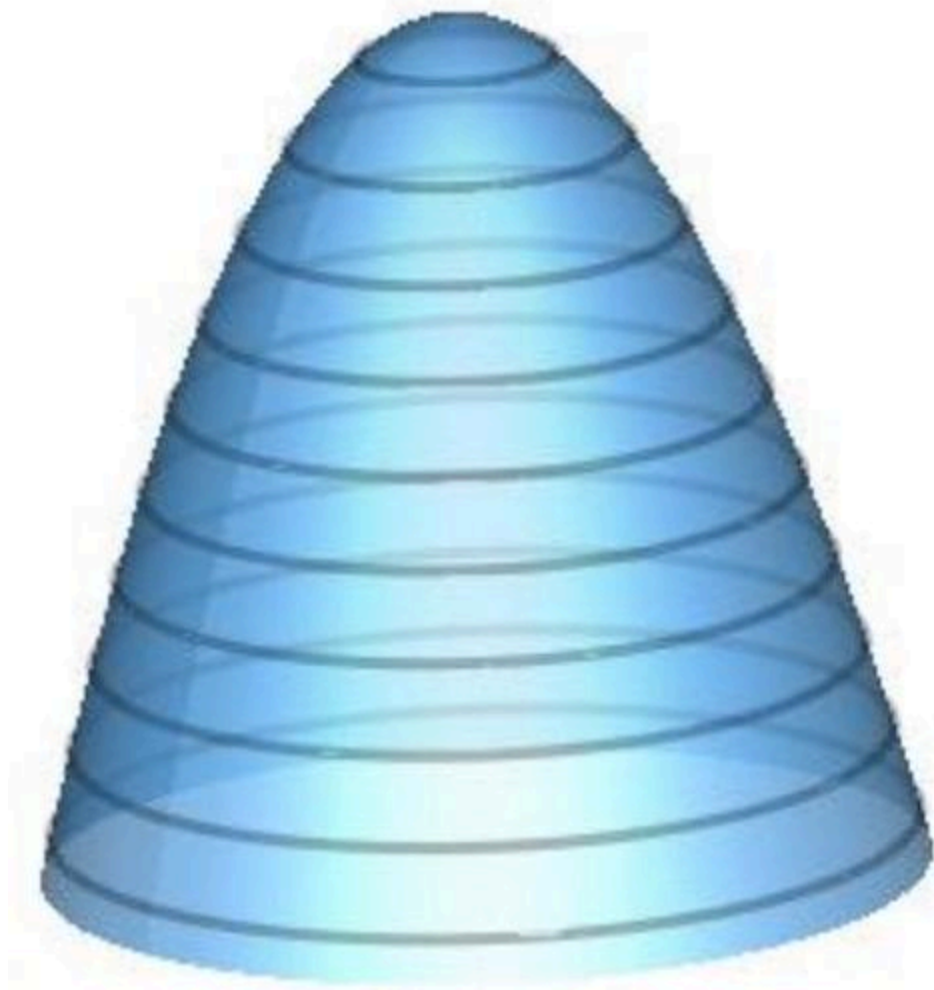




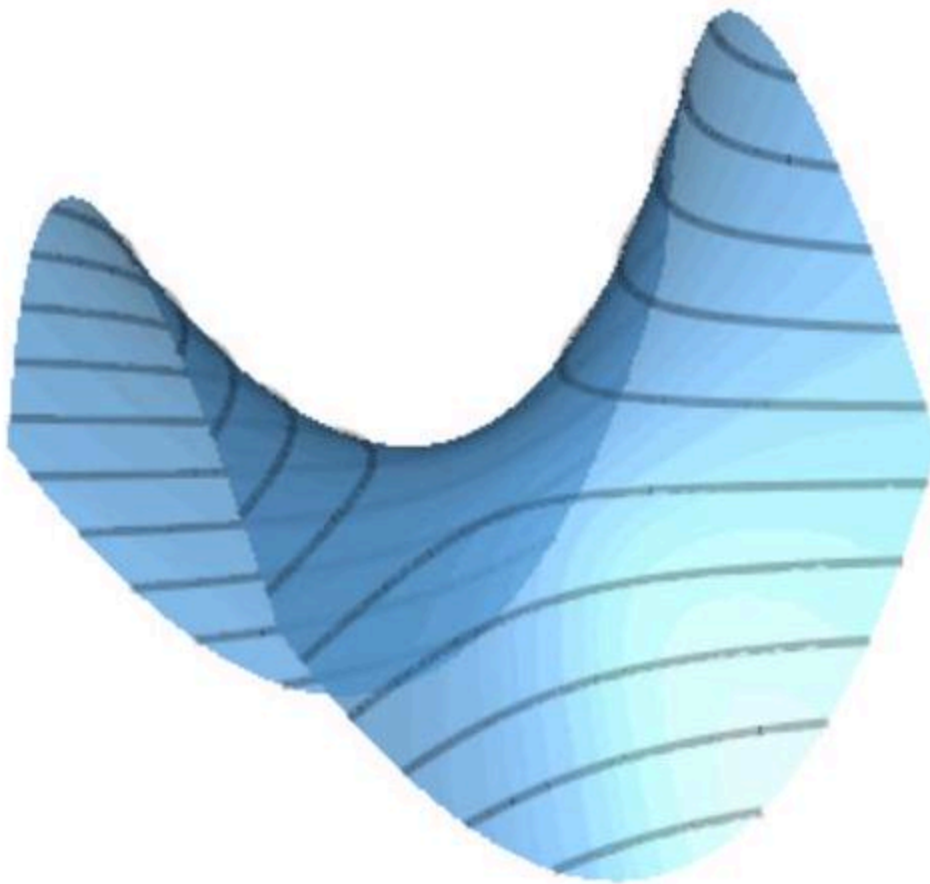




Concentric circles (ellipses): minimum



Concentric circles (ellipses): maximum

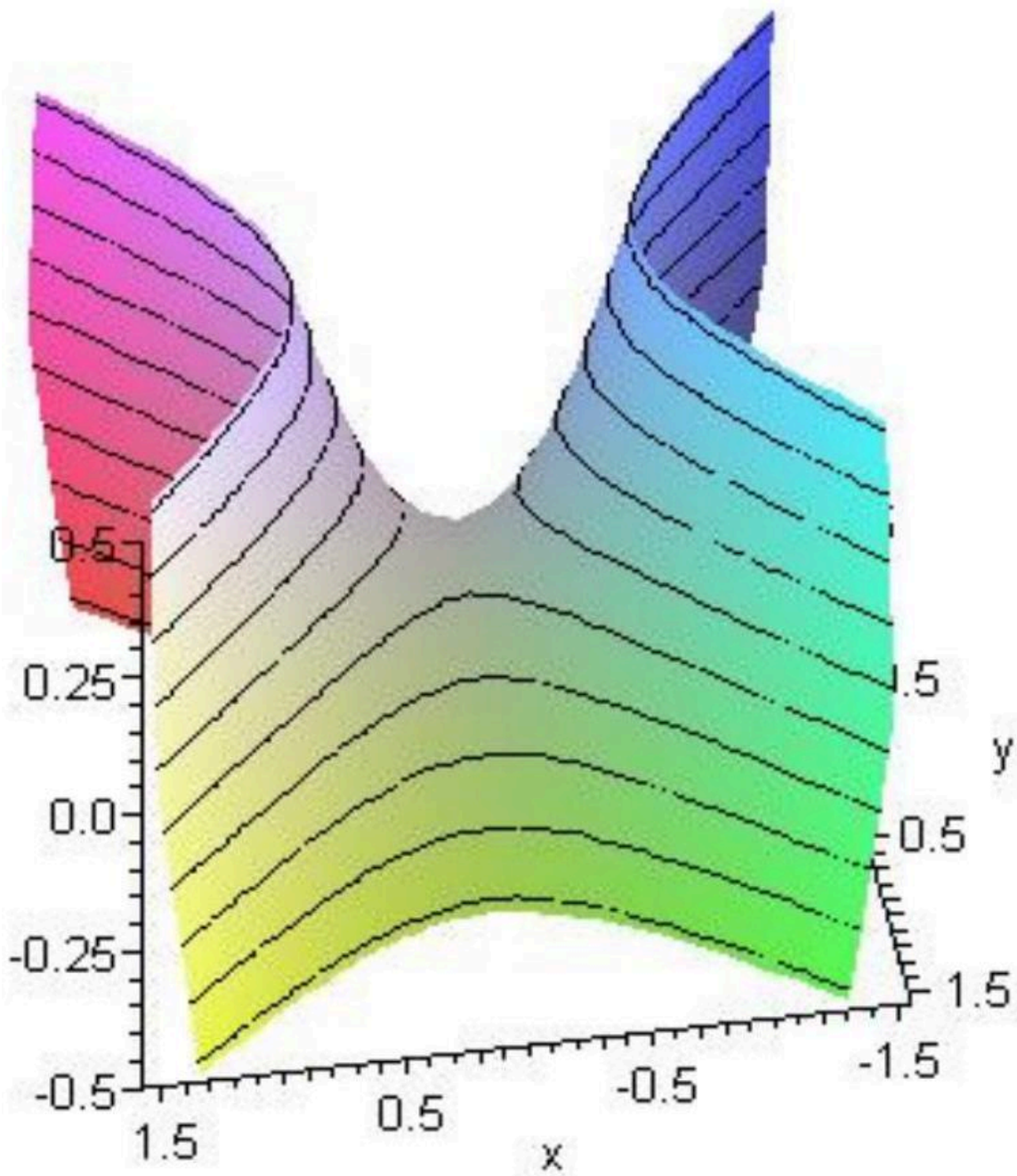


Hyperbolas: saddle point

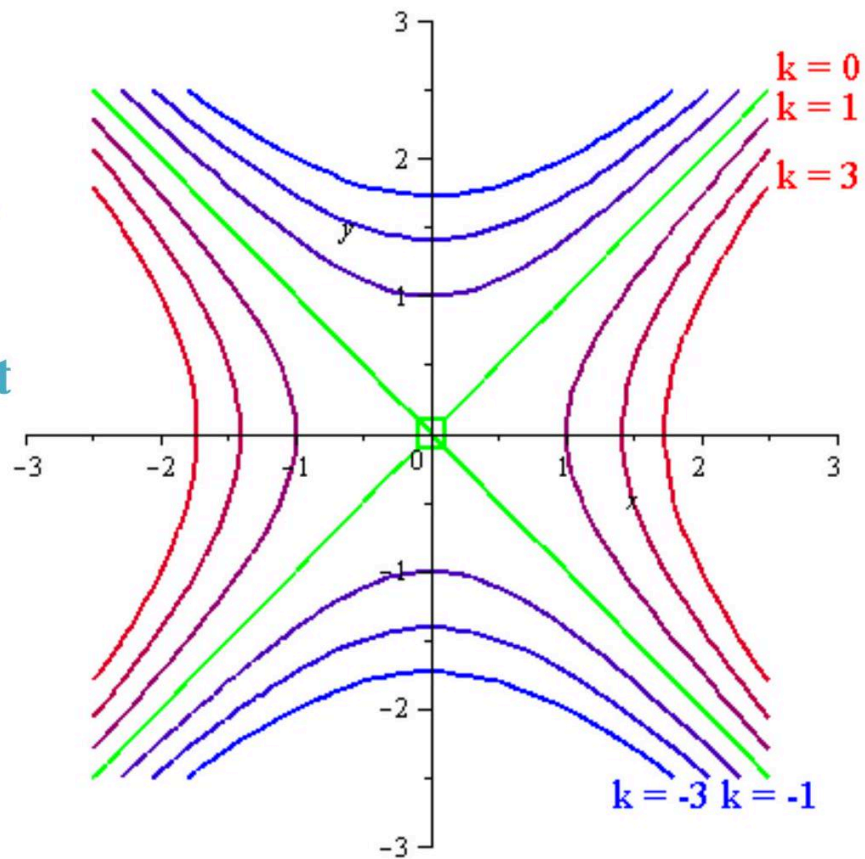
Closely spaced contours: steep slope in 3D. Widely spaced contours: flat 3D surface.

Functions of Several Variables

Hyperbolic paraboloid (Saddle) $z = x^2 - y^2$



Level curves: $x^2 - y^2 = k$



$k = 0$ gives the asymptotes: $y = x$ and $y = -x$.

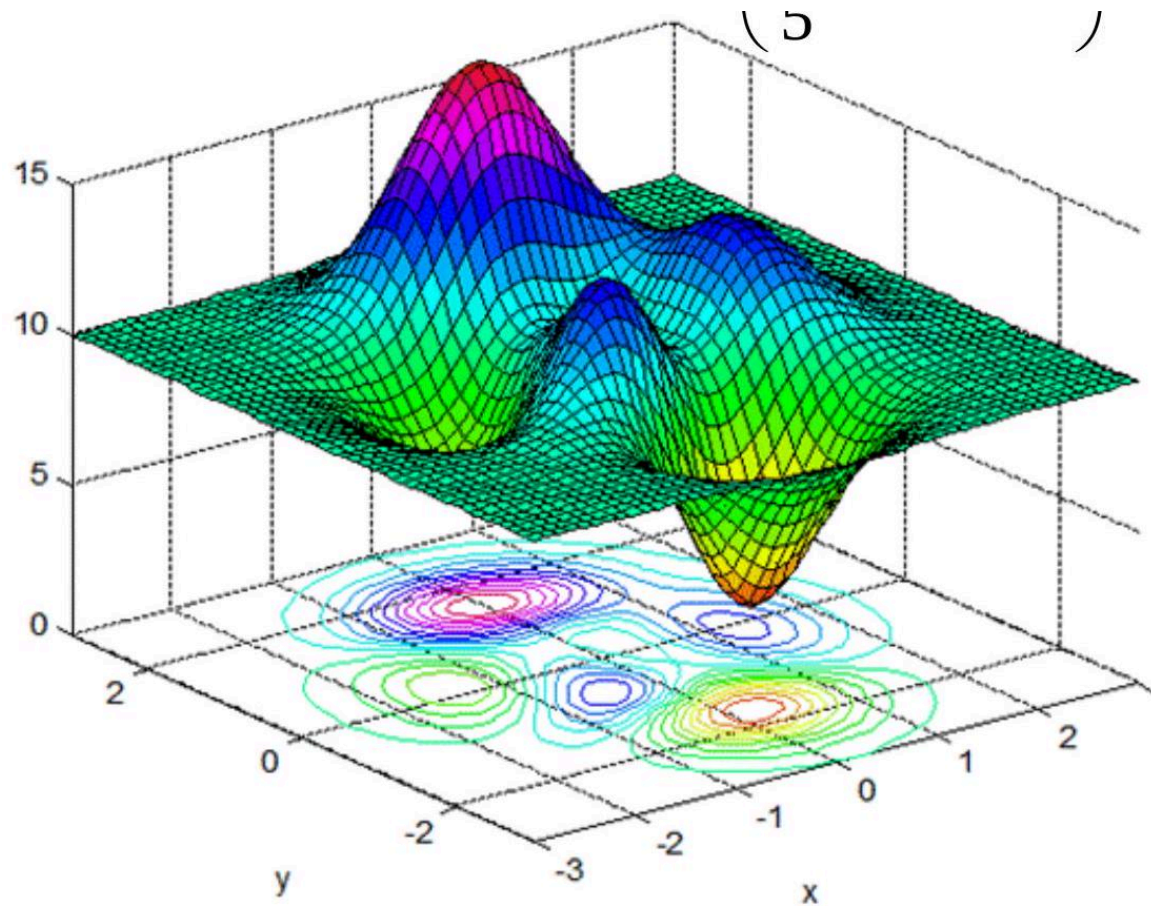
For all the other values of k we get hyperbolas.

Note that the values of k decrease as we approach the origin from the x -axis and they increase as we approach the origin from the y -axis.

Functions of Several Variables

Use the contour plot to identify maximum, minimum and saddle points of the function

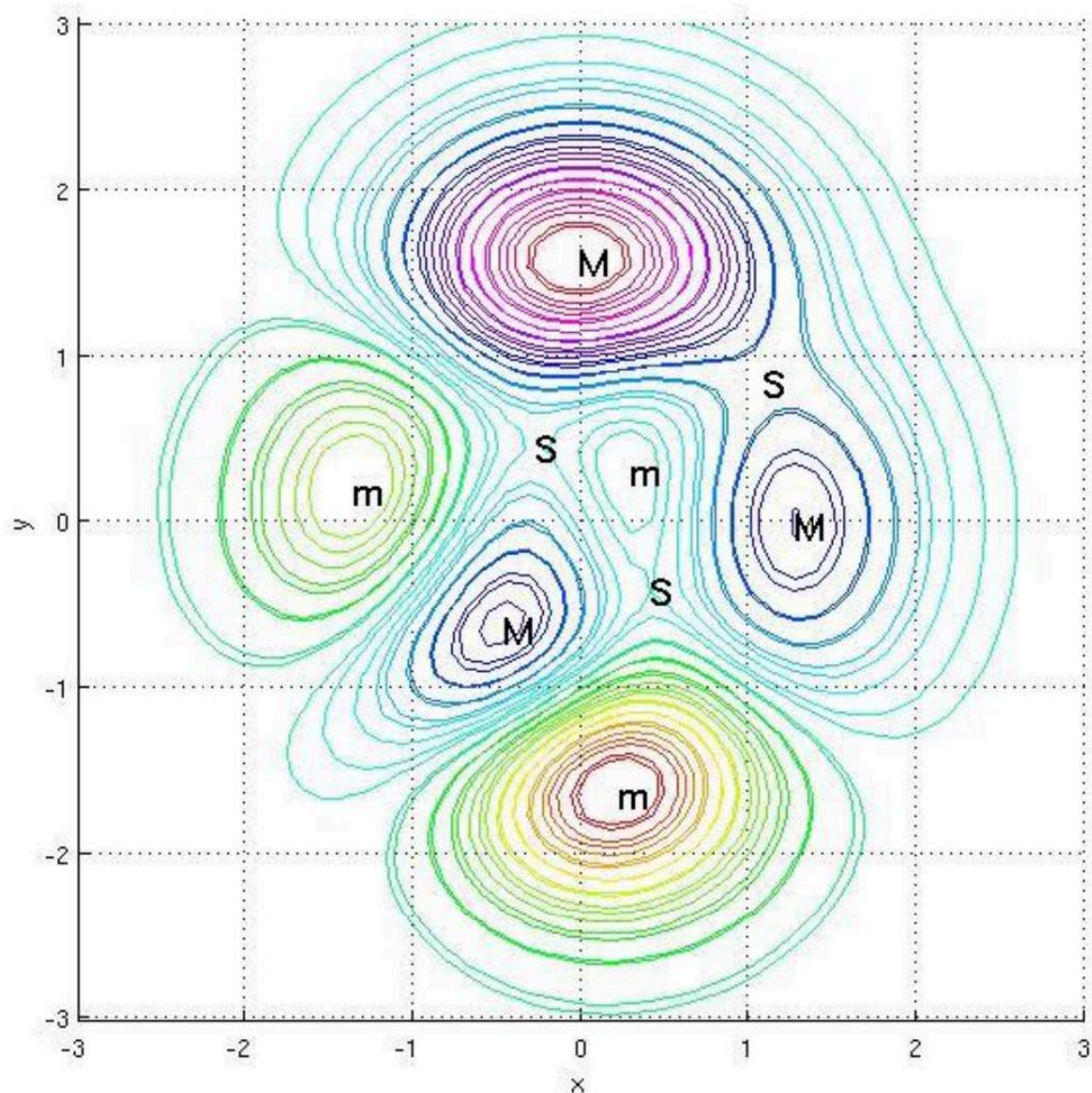
$$z = 3(1 - x)^2 e^{-x^2 - (y+1)^2} - 10 \left(\frac{x}{5} - x^3 - y^5 \right) e^{-x^2 - y^2} - \frac{1}{3} e^{-(x+1)^2 - y^2} + 10$$



Local maxima:

$$(1.3, -0.005), (-0.46, -0.63), (-0.01, 1.6)$$

Local minima:



$(.23,-1.6), (-1.3,.20), (.3,.3)$

Saddle points:

$(-.27,.47), (.42,-.39), (1.1,.85)$

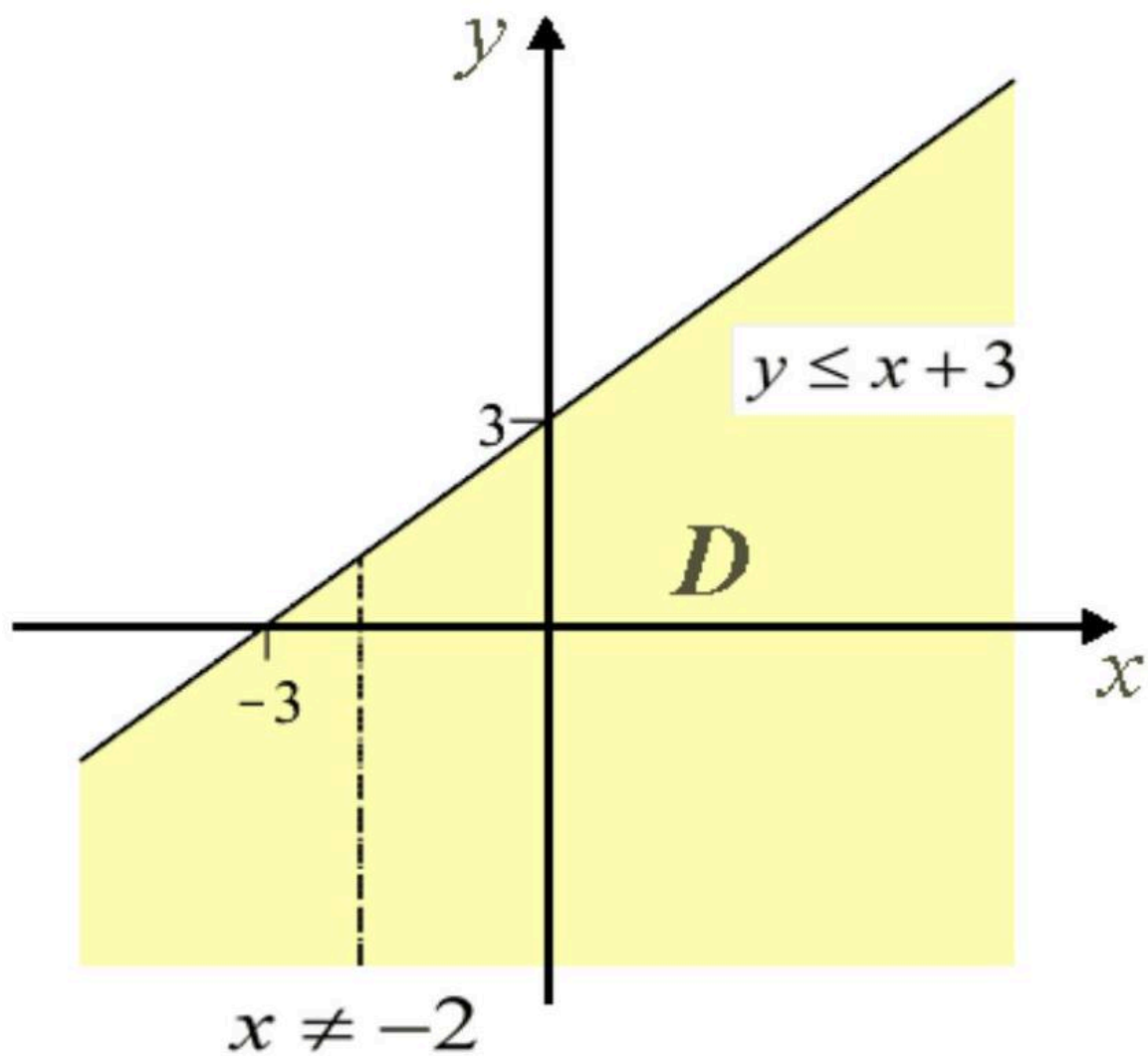
Functions of Several Variables

Unless otherwise specified, the Domain of the function $f(x, y)$ is the set of all points (x, y) for which $f(x, y)$ is a well defined real number.

The Range of the function $f(x, y)$ is the set $\{z \mid z = f(x, y) \text{ with } (x, y) \in \text{Domain}\}$

Example: Find the domain: $f(x, y) = \frac{\sqrt{x-y+3}}{x+2}$

The function is defined when $x - y + 3 \geq 0$ and $x + 2 \neq 0$

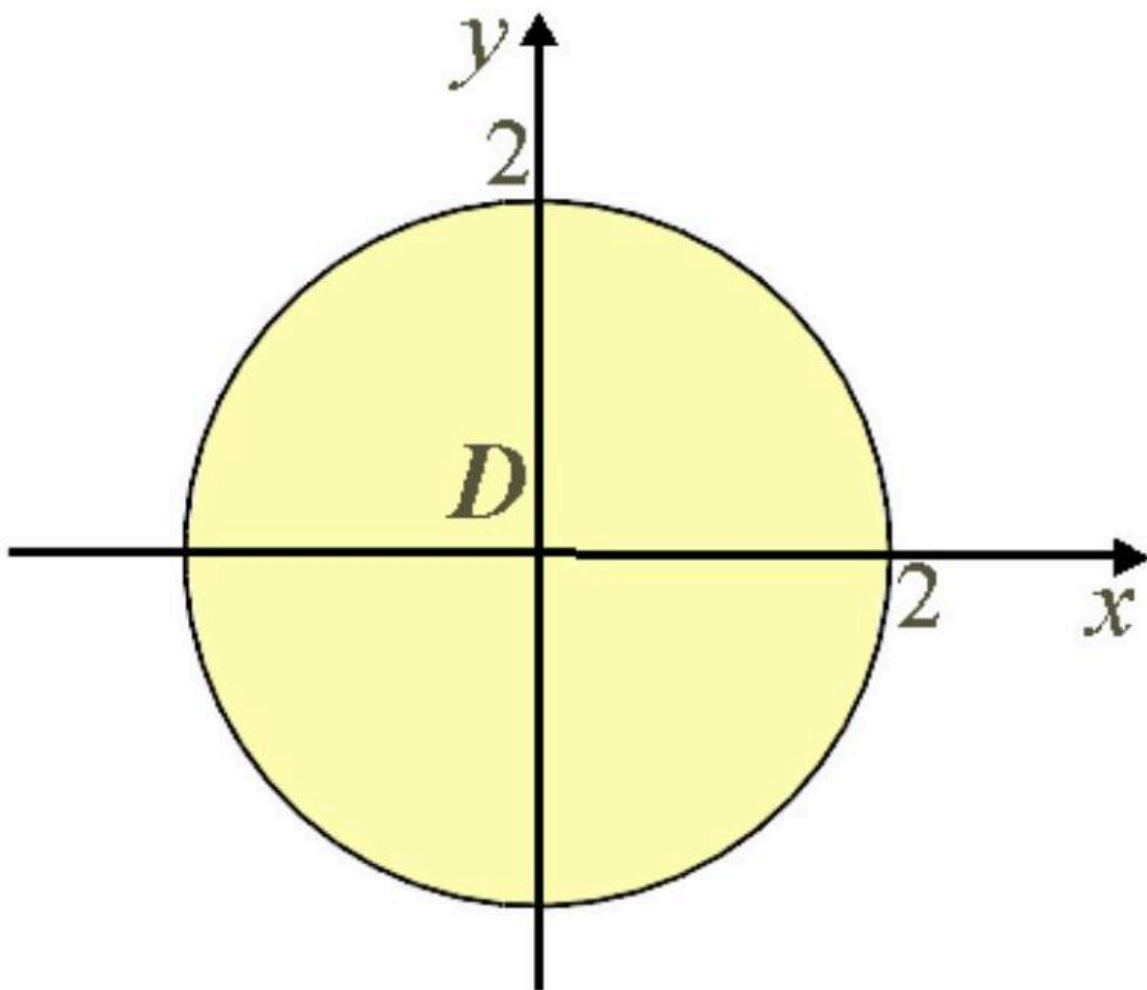


Functions of Several Variables

Example: Find domain and range: $f(x, y) = \sqrt{4 - x^2 - y^2}$

Domain:

$$4 - x^2 - y^2 \geq 0 \Rightarrow x^2 + y^2 \leq 4$$



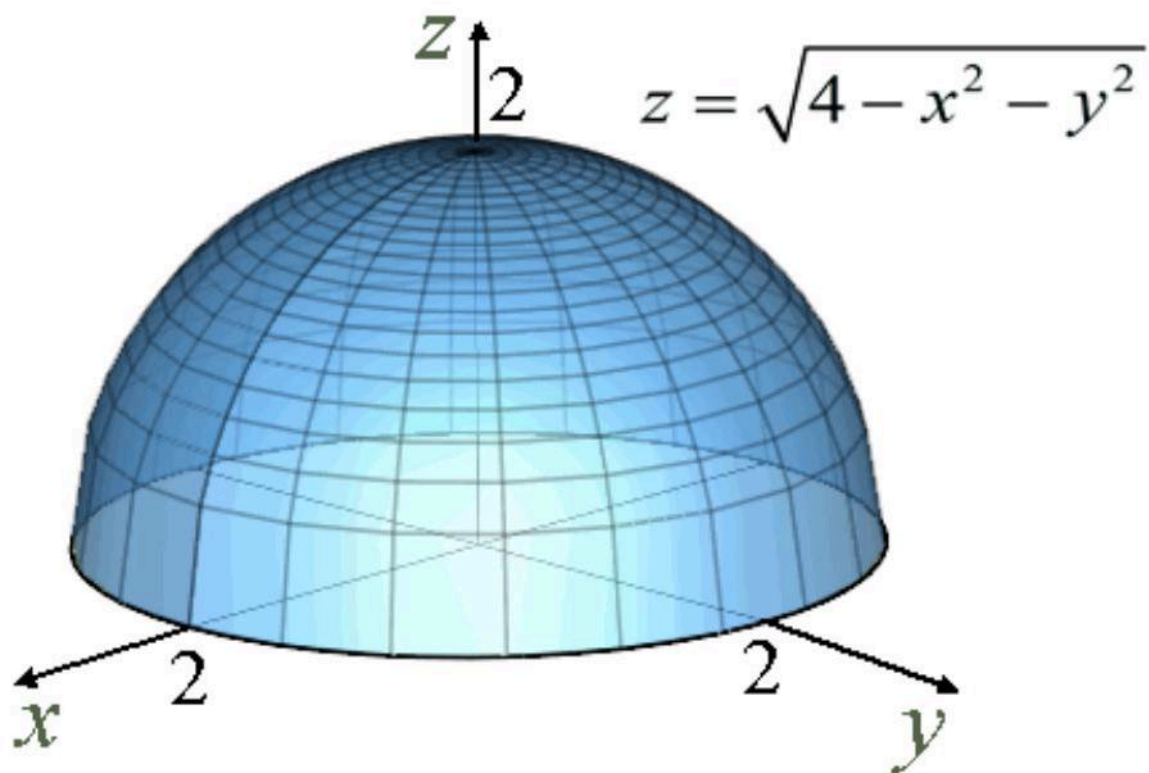
The surface is the top half of the sphere of radius 2 :

$$x^2 + y^2 + z^2 = 4$$

Range:

The largest value of z is $\sqrt{4} = 2$, obtained when $x = 0$ and $y = 0$.

On the other hand, the square root is always non negative, thus $z \geq 0$ and the range is the interval $[0, 2]$



Function of Several Variables

A function of n variables assigns a number $z = f(x_1, x_2, \dots, x_n)$ to an n -tuple $(x_1, x_2, \dots, x_n)$

Example: Consider the function of three variable $f(x, y, z) = x^2 \ln(x - y + z)$

a) Evaluate $f(3, 6, 4)$

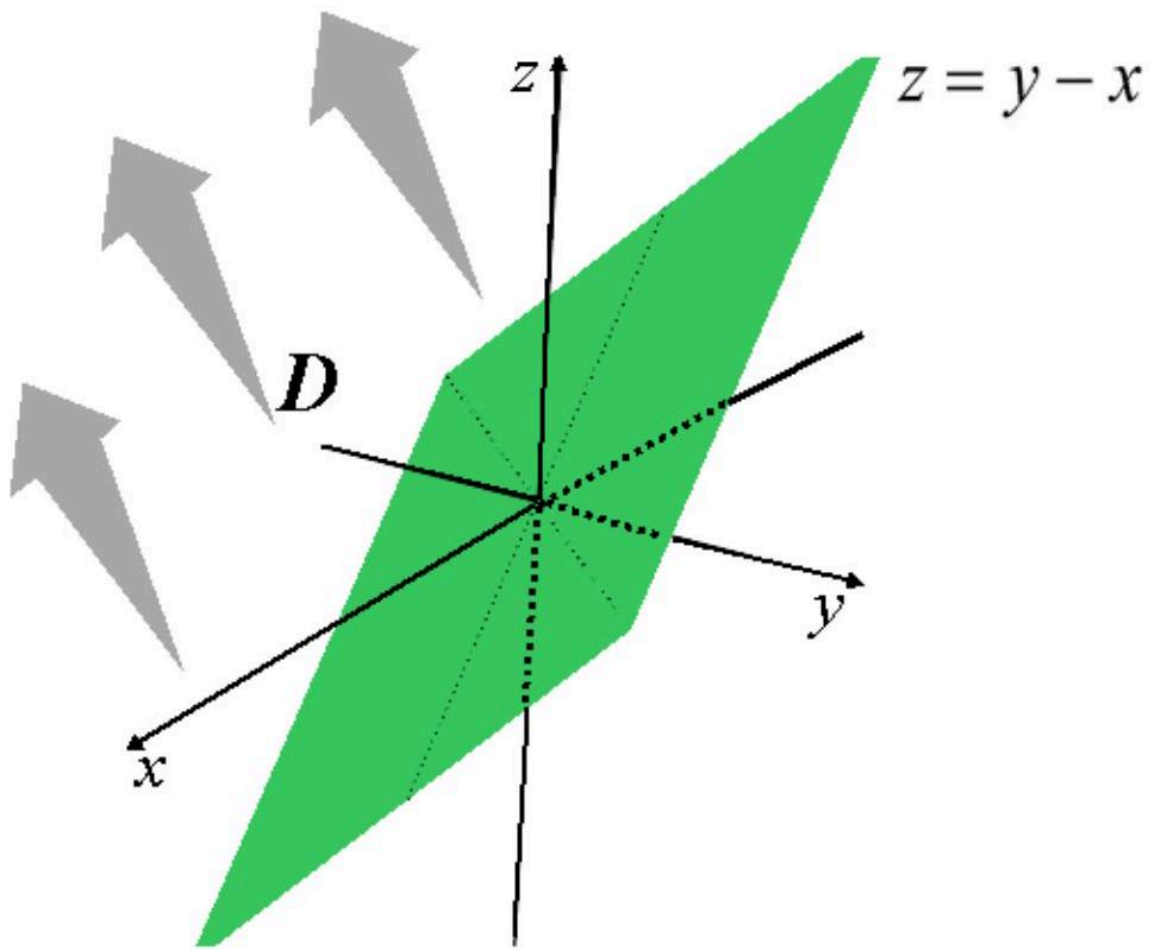
$$f(3, 6, 4) = 3^2 \ln(3 - 6 + 4) = 9 \ln(1) = 0$$

b) Find the domain of f

$$D = \{(x, y, z) \mid x - y + z > 0\}$$

The plane $x - y + z = 0$ divides the space in two parts. The part of space above the plane constitutes the domain of the function.

c) Find the range of f



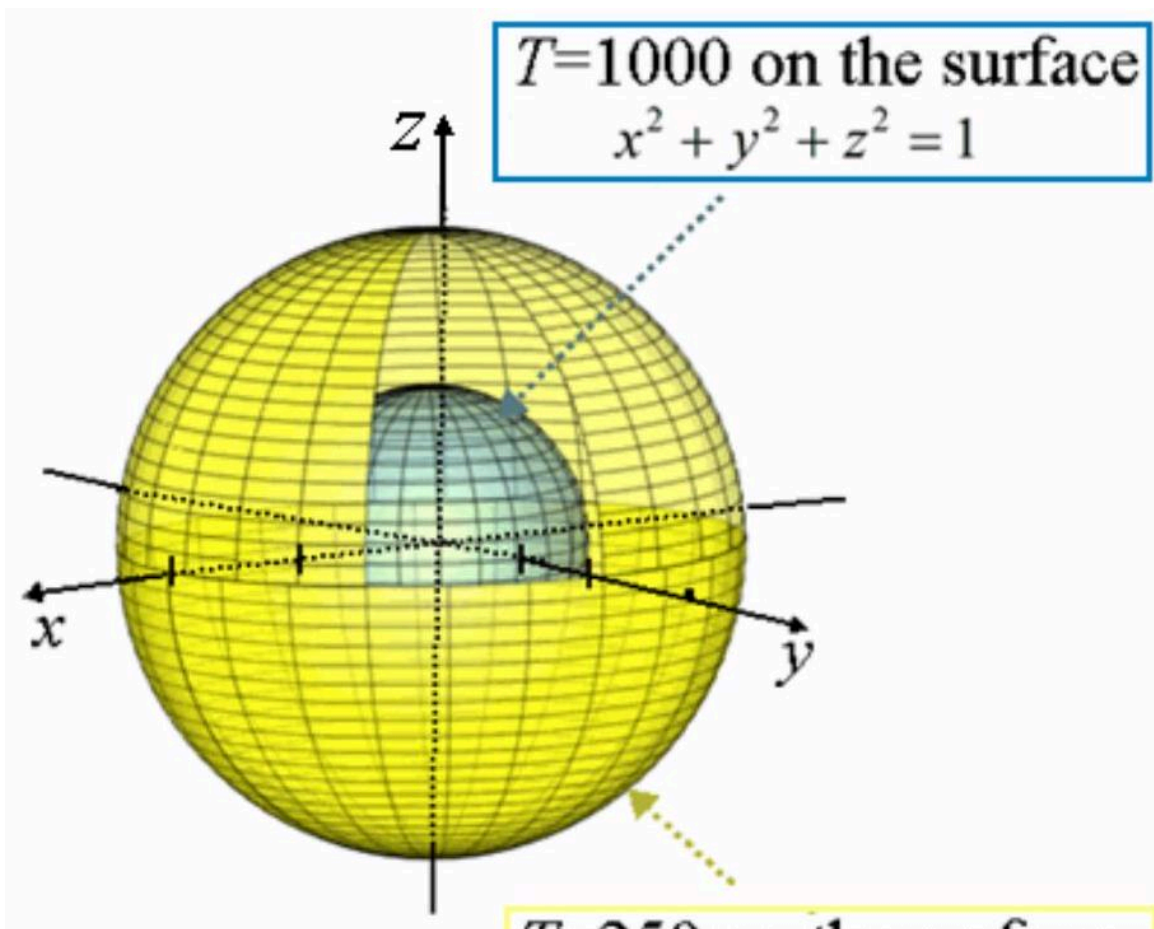
Since the natural logarithmic function takes on all possible real values, the domain of the function is $(-\infty, \infty)$

Function of Several Variables

Example: The temperature in a region where there is a heat source at the origin is given by

$$T = \frac{1000}{x^2 + y^2 + z^2} \text{ for } x^2 + y^2 + z^2 \geq 1$$

We cannot graph this "surface" because we would need four dimensions, but if we set $T(x, y, z) = k$ with k constant, we obtain level surfaces which in this context are isotherms.



$T = 250$ on the surface
 $x^2 + y^2 + z^2 = 4$

$$\frac{1000}{x^2 + y^2 + z^2} = k$$

We can pick arbitrary values for k .
 Let's pick $k = 1000$ and $k = 250$

$$k = 1000 \Rightarrow \frac{1000}{x^2 + y^2 + z^2} = 1000 \Rightarrow x^2 + y^2 + z^2 = 1$$

$$k = 250 \Rightarrow \frac{1000}{x^2 + y^2 + z^2} = 250 \Rightarrow x^2 + y^2 + z^2 = 4$$